



P.E.S. College of Engineering, Mandya-571401

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Department of Information Science and Engineering



Kannada Speech Emotion Recognition

Team Members:

- Parikshith S – 4PS22IS039
- Pavana S – 4PS22IS040
- Sahana S Gudennavar – 4PS22IS045
- Varshini M R – 4PS22IS056

Under the Guidance of:

Sushmitha S.R
Assistant Professor
Dept. Of IS&E

Table of Contents

Sl.NO	Content Title
1.	Introduction
2.	Problem Statement
3.	Objectives 1 Methodology
4.	Objectives 2 Methodology
5.	Objectives 3 Methodology
6.	Output
7.	Tools and Technologies
8.	References

Introduction

- Speech contains emotional information beyond words
- Most SER systems focus only on English
- Kannada language lacks advanced emotion recognition systems
- Need for real-time emotion detection system

SDG Contribution:

SDG 9: Industry, Innovation, and Infrastructure

→ Develops an intelligent AI-based system for Kannada speech emotion detection

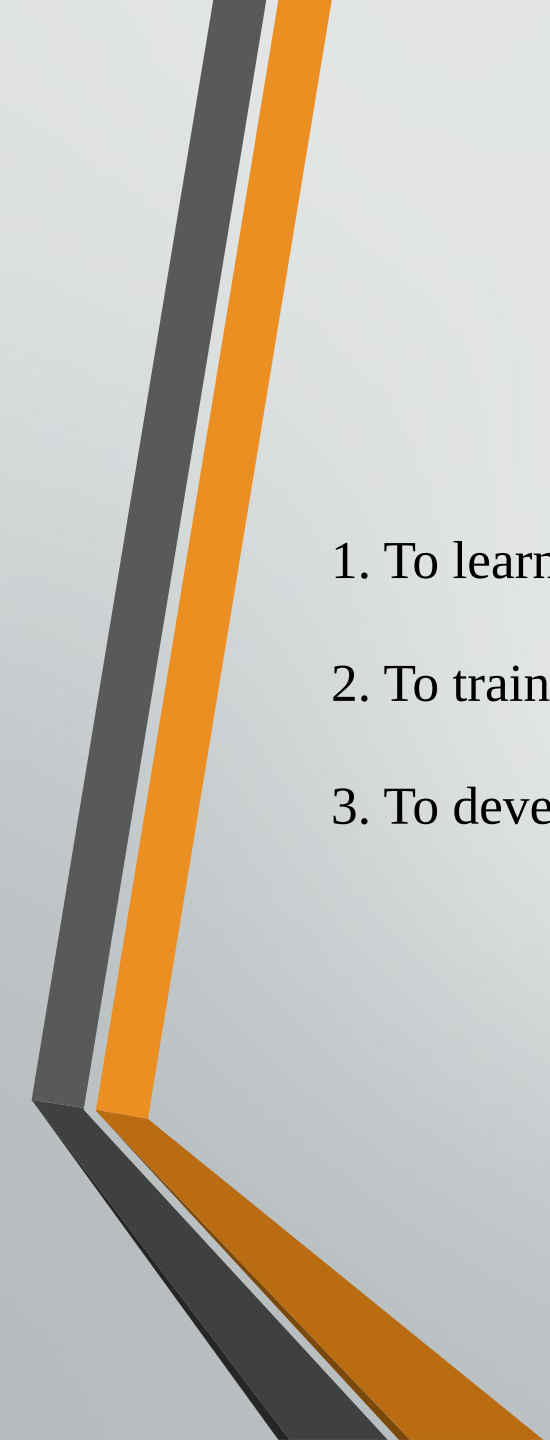
SDG 3: Good Health and Well-being

→ Helps in mental health monitoring and emotional analysis

Problem Statement:

Existing Speech Emotion Recognition systems are mainly designed for high-resource languages like English and perform poorly on Kannada speech due to lack of datasets and linguistic differences . These systems rely on traditional machine learning methods and fail to capture complex emotional patterns in speech . Therefore, an efficient system is required to accurately recognize emotions from Kannada speech using advanced deep learning techniques and provide real-time predictions





Objectives

1. To learn deep acoustic features from Kannada speech using transformer-based models
2. To train and fine-tune a deep learning model for accurate emotion classification
3. To develop a real-time full-stack system for speech emotion recognition

Objective 1

Goal:

- Learn meaningful speech features from raw Kannada audio

Methodology:

- Input raw speech audio
- Process audio at 16kHz
- Extract acoustic patterns (pitch, energy, MFCC)
- Use deep feature learning approach

Implementation:

- Used Kannada speech dataset
- Applied preprocessing and augmentation
- Generated structured audio inputs for model

Objective 2

Goal:

- Train a robust deep learning model for emotion classification

Methodology:

- Use Wav2Vec2 (Transformer model)
- Fine-tune on Kannada dataset
- Learn speech representations automatically
- Perform multi-class classification (5 emotions)

Implementation:

- Trained model using PyTorch
- Used GPU for training
- Achieved 84% accuracy and 0.84 F1-score

Objective 3

Goal:

- Develop a real-time speech emotion recognition system

Methodology:

- Capture input via microphone or file upload
- Send audio to backend
- Perform model inference
- Display predicted emotion

Implementation:

- Frontend: React + TypeScript
- Backend: Flask API
- Database: MongoDB
- Integrated full pipeline (Input → Prediction → Output)

Output

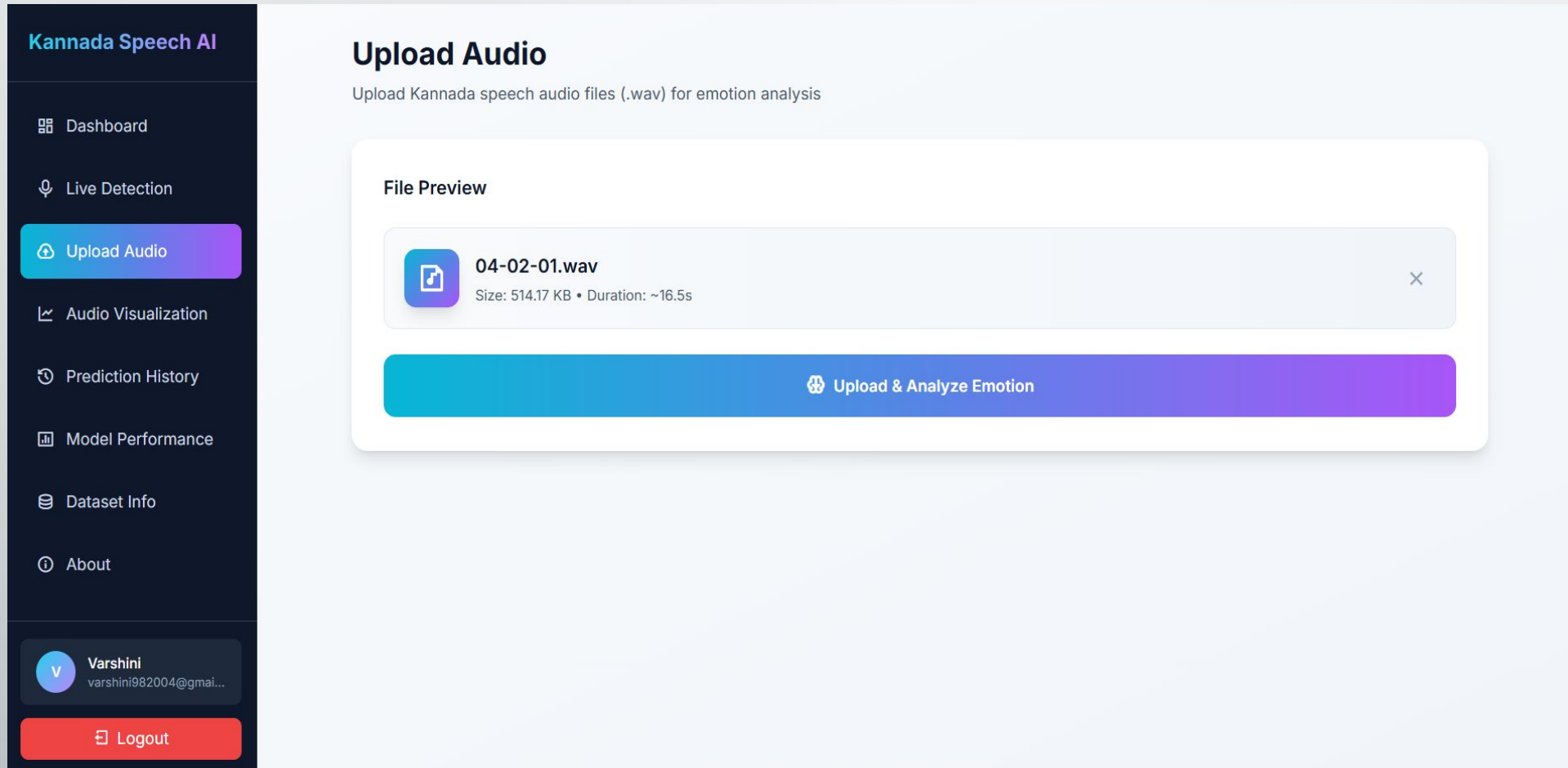


Fig 1: Audio Upload / Live Detection Interface

Output

The screenshot displays the 'Kannada Speech AI' web application interface. On the left is a dark sidebar with navigation links: Dashboard, Live Detection, Upload Audio (highlighted in purple), Audio Visualization, Prediction History, Model Performance, Dataset Info, and About. At the bottom of the sidebar is a user profile for 'Varshini' and a red 'Logout' button. The main content area is titled 'Upload Audio' with a subtitle 'Upload Kannada speech audio files (.wav) for emotion analysis'. It features a 'File Preview' section showing an uploaded file '04-02-01.wav' with a size of 514.17 KB and a duration of ~16.5s. Below this is a 'Processing Pipeline' with four steps: 'Uploading to Cloudinary' (completed with a checkmark), 'Processing Audio' (active with a circular arrow), 'Extracting Features' (step 3), and 'Classifying Emotion' (step 4).

Kannada Speech AI

- Dashboard
- Live Detection
- Upload Audio**
- Audio Visualization
- Prediction History
- Model Performance
- Dataset Info
- About

Varshini
varshini982004@gmail...

Logout

Upload Audio

Upload Kannada speech audio files (.wav) for emotion analysis

File Preview

04-02-01.wav
Size: 514.17 KB • Duration: ~16.5s

Processing Pipeline

- Uploading to Cloudinary
- Processing Audio
- Extracting Features
- Classifying Emotion

Fig 2: Model Inference Process

Output



Fig 3: Emotion Prediction Output

Output

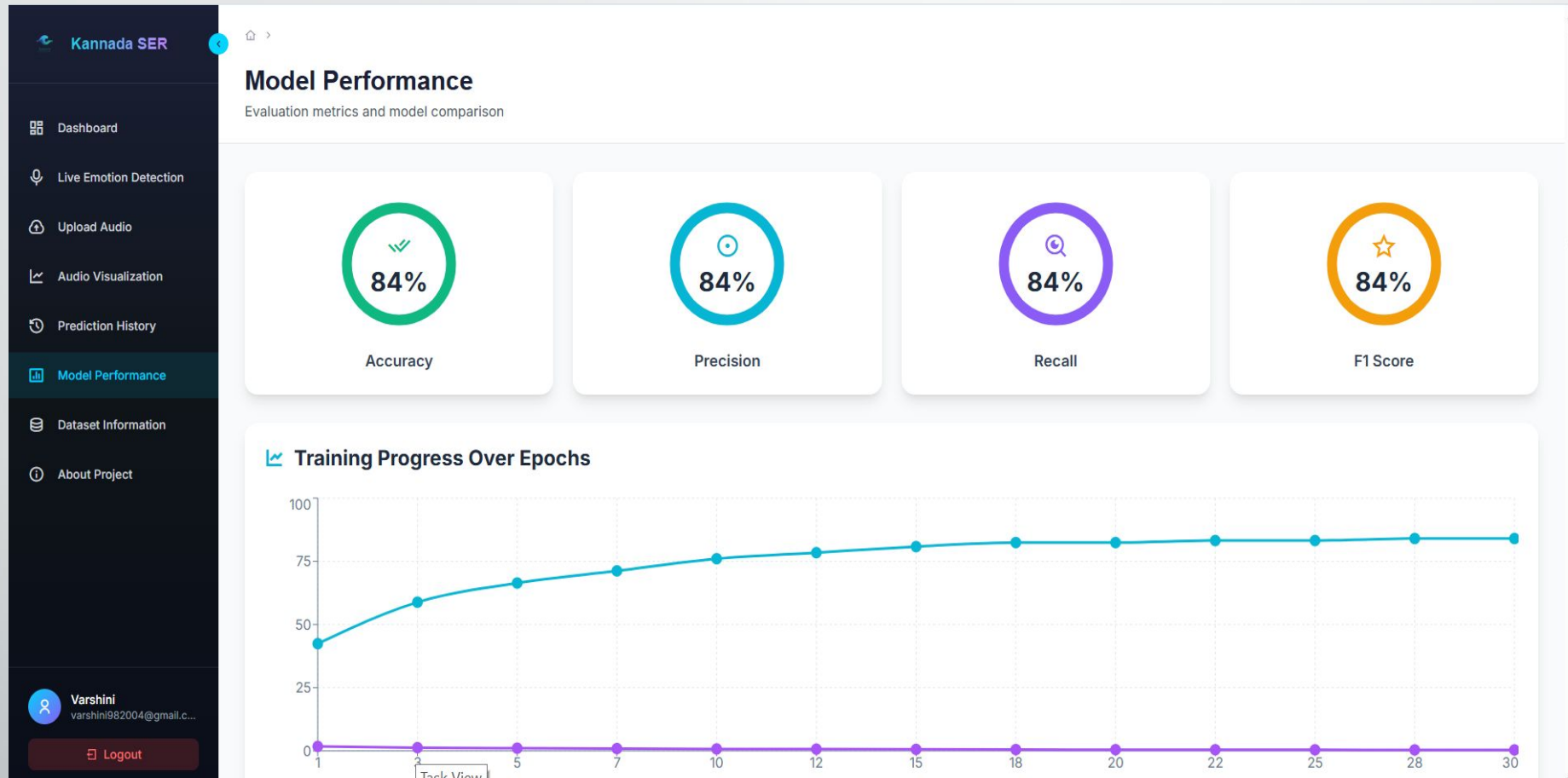


Fig 4.Model Performance

Tools and Technologies

Programming & Frameworks

- Python
- React.js
- Flask

Deep Learning & Libraries

- PyTorch
- Transformers (Wav2Vec2)
- Librosa

Model & Techniques

- Wav2Vec2 Transformer – based model
- Dual-model inference
- Feature extraction (MFCC, Pitch, RMS)

Development

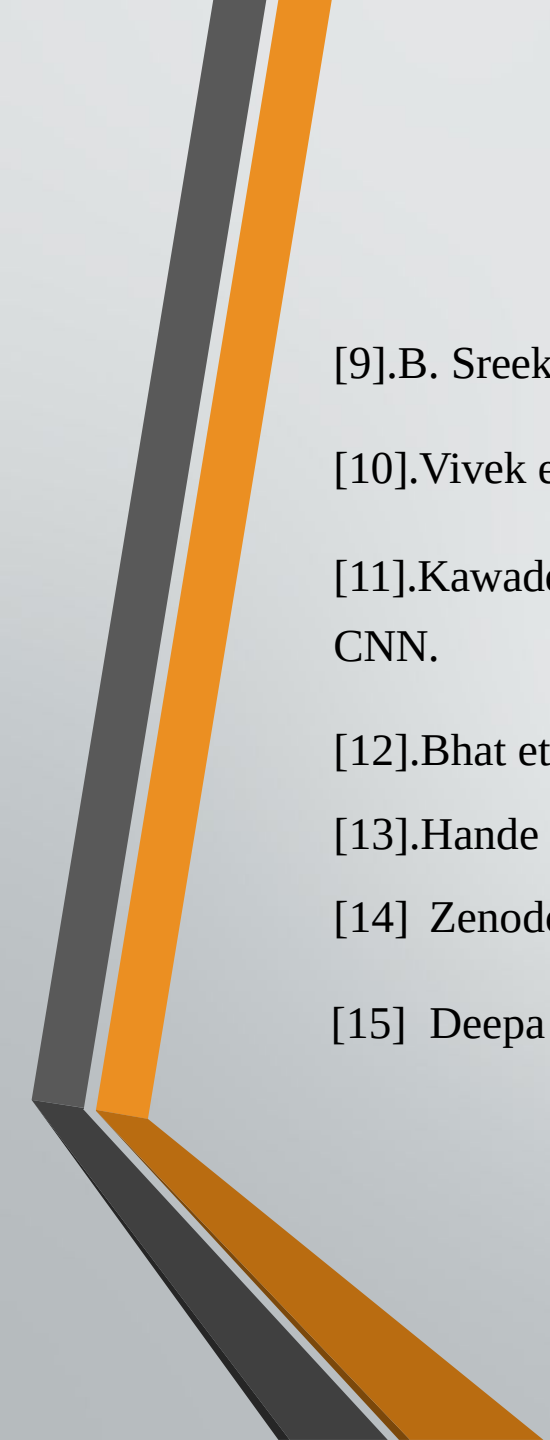
- GPU (training)
- CPU (deployment)

Dataset

- Hybrid Dataset: Zonoda + Custom Kannada Speech dataset

References

- [1].IEEE Authors (Multiple) (2021). Speech Emotion Recognition using CNN.
- [2].Bhattacharya T. et al. (IEEE) (2024). Deep Learning Approaches for Speech Emotion Recognition.
- [3].Li X. et al. (Elsevier) (2024). Transformer Models for Multilingual Emotion Recognition.
- [4].Kumar R., Singh A. (Springer) (2022). Emotion Recognition in RAVDESS Dataset using SVM.
- [5]. Han K., Yu D. (Elsevier) (2023). Speech Emotion Analysis using R.
- [6].Harshith B U. et al. (IERT) (2019). Speaker Dependent Emotion Recognition from Speech for Kannada Language.
- [7].Pallavi J., Geethashree A., D J Ravi (2014). Emotional Analysis and Evaluation of Kannada Speech Database.
- [8].Smrithi Baliga et al. (2023). Kannada Speech Emotion Recognition Using Ensembling Techniques.



[9].B. Sreekantha et al. (2023). LSTM Based Kannada Speech Emotion Recognition.

[10].Vivek et al. (2023). Lightweight SER for Regional Languages (Kannada Case Study).

[11].Kawade & Jagtap (RIA) (2024). Indian Cross-Corpus SER using Spectral-Temporal-Voice Features and Deep CNN.

[12].Bhat et. al. (IJCA) (2024). Acoustic Feature Mapping for Kannada Emotion Recognition.

[13].Hande et al. (arXiv) (2021). Hope Speech Detection in Under-resourced Kannada Language.

[14] Zenodo Dataset Authors (2022). Kannada Emotional Speech Dataset (Open Database).

[15] Deepa et al. (IEEE) (2023). Transformer-based Hybrid Model for Multilingual SER.



THANK YOU